VRAJ PATEL

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TECHNICAL SKILLS

Languages	C++, C, Python, Java, x86 Assembly
Databases	SQL, PostgreSQL
Backend	RESTful API Integration
Web Technologies	HTML, CSS, Bootstrap
Version Control	Git
Systems & Networking	Linux (Ubuntu), Windows
Tools	Jira, Confluence
Development Tools:	Visual Studio Code, Android Studio
Testing:	GoogleTest, Selenium
Core Concepts	Data Structures and Algorithms, Object-Oriented Programming (OOP), Debugging, Code Optimization, SDLC, Agile Methodology

EDUCATION

BACHELOR OF SCIENCE – COMPUTING SCIENCE — **Burnaby, BC**
Simon Fraser University — *June 2025*

- Concentration: **Programming Languages and Software**
- Minor: **Mathematics**
- Relevant Coursework**: Computer Systems, Operating Systems, Networking, Database Systems, Software Engineering**
- GPA**:** 2.90

WORK EXPERIENCE

WEB DEVELOPER — **Burnaby, BC**
CJSF 90.1FM — *January 2024 – May 2024*

- Designed and implemented responsive web components using HTML, JavaScript, and CSS, improving UI consistency across devices, reducing layout issues by approximately 15%.
- Converted Figma mockups into responsive production web pages and maintained Drupal CMS content, supporting consistent publishing processes and accessibility standards across 10+ pages.
- Automated livestream setup by integrating Stream Deck with OBS Studio, eliminating manual broadcast steps.

TECHNICAL WRITER INTERN — **Waterloo, ON**
BlackBerry (IVY and Cybersecurity) — *September 2022 – April 2023*

- Implemented C++ fixes for SDK defects found during API validation and submitted patches through code review, reducing recurring issue reports by around 20%.
- Replaced manual Excel ingestion with a server-hosted SQL pipeline, reducing processing time by 30% and enabling scalable data integration into Adobe Experience Manager.
- Validated backend APIs and SDK components in a Linux environment using C++ and Python to debug issues and ensure documentation accuracy.
- Built Python assertion-based test scripts to catch incorrect API behavior early, increasing automated test coverage and reducing manual validation effort.
- Authored 30+ XML validation rules (Schematron) that enforced documentation standards and reduced publishing inconsistencies by roughly 40%.
- Applied frontend fixes in Adobe Experience Manager to address WCAG accessibility compliance findings.

PROJECTS

MULTI-CLIENT TCP CHAT SERVER (View on GitHub)

Technologies used: C, Pthreads, Linux, TCP/IP

- Built a multi-threaded TCP server in C that handles concurrent client connections and broadcasts messages.
- Used pthreads for concurrency and implemented basic synchronization to ensure correct message handling.
- Implemented a simple message protocol that includes sender IP/port information and supports client termination.
- Developed concurrent client programs to simulate network traffic and validate synchronization, thread behaviour and socket communication.

HEART DISEASE PREDICTION MODEL (View on GitHub)

Technologies used: Python, Numpy, Pandas, Matplotlib, Scikit-Learn, Statsmodels, Jupyter Notebook

- Developed a binary classification model to predict heart disease risk using clinical and demographic indicators from a large-scale public health survey dataset.
- Prepared the dataset by cleaning records, converting categorical variables into numeric values, and removing outliers before model training.
- Trained and benchmarked multiple machine learning models (Linear Regression, Random Forest, KNN, XGBoost) using an 80/20 train-test split, with hyperparameter tuning.
- Achieved 91.4% accuracy and evaluated precision/recall trade-offs to address imbalanced data.

GAMEINN - Full-Stack Gaming Platform (View on GitHub)

Technologies used: HTML, CSS, JavaScript, Bootstrap, Java

- Designed a Java backend with PostgreSQL to manage user authentication, group matching logic, and database transactions.
- Developed backend logic for a group finder feature, allowing users to create and join gaming queues based on shared preferences.
- Integrated the IGDB API to fetch and display top game titles and built a searchable clip-sharing page for users to post and discover gaming content.

SPARSE MATRIX PERFORMANCE EVALUATION (View on GitHub)

Technologies used: C/C++

- Benchmarked sparse matrix operations across L1/L2/L3 cache levels under different matrix sizes and compiler optimization levels, showing slowdown as data exceeded CPU cache capacity.
- Applied SIMD optimizations, achieving roughly 3-4× speedup on large matrix operations.
- Profiled with perf and Cachegrind to analyze cache misses, branch behavior, and memory locality, observing >95% L1 cache misses during multiplication.

TEACHING EXPERIENCE

VOLUNTEER TEACHING ASSISTANT – Python Programming **Remote**

Public Policy in Africa Initiative (PPiAI) *January 2021 – December 2022*

Published Book | ***Sample Lecture***

- Supported remote Python programming sessions alongside a lead data engineer, guiding students through coding exercises, debugging techniques, and core programming concepts.
- Contributed to preparing and organizing lecture materials later published as a textbook.
- Provided one-on-one support to students during live sessions, reinforcing problem-solving and programming fundamentals.

ADDITIONAL EXPERIENCE

RETAIL TEAM LEAD – PART TIME **Burnaby, BC**

Mountain Warehouse, Metropolis *July 2023 – Present*

- Supervised daily store operations, coordinating associates and stock handling to ensure smooth and efficient floor execution.
- Managed product levels and organized replenishment to maintain accurate and consistent stock availability.